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CS3650

Lab 2

Prof. Mallapuram

NOTES:

Log files are attached aswell, I think it missed some of them as I had a crash (other process on the computer caused), [GitHub](#) will have backup.

Step 1:

I updated the system packages using
`sudo apt update`.

```
hdoop@ograzier:~$ sudo apt update && sudo apt install
[sudo: authenticate] Password:
Hit:1 http://us.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:3 http://us.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 http://us.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages (616 kB)
Hit:5 http://security.ubuntu.com/ubuntu resolute-security InRelease
Get:6 http://us.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [294 kB]
Get:7 https://pkgs.tailscale.com/stable/ubuntu resolute InRelease
Fetched 1,054 kB in 0s (2,431 kB/s)
39 packages can be upgraded. Run 'apt list --upgradable' to see them.
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 39
```

Step 2:

I installed OpenJDK 8 using `sudo apt install openjdk-8-jdk` and verified the version using `java -version`. (Outdated -- Ended up swapping to openjdk-17-jdk)

```
hdoop@ograzier:~$ java -version
openjdk version "17.0.20" 2026-07-21
OpenJDK Runtime Environment (build 17.0.20+8-1-26.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 17.0.20+8-1-26.04-Ubuntu, mixed mode, sharing)
```

Step 3:

I downloaded the latest release of Hadoop (Hadoop 3.5.0) via `wget`, extracted the tar package, and created the required directory structures for the NameNode, DataNode, and temporary data (`tmpdata`, `dfsdata/namenode`, `dfsdata/datanode`).

Step 4:

I created a dedicated non-root user called `hdoop` with `sudo` privileges and set up passwordless SSH on `localhost` as required.

```
hdoop@ograzier:~$ ssh localhost
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

35 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

14 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Wed Sep 16 11:14:52 2026 from 127.0.0.1
```

I initially set up SSH and downloaded Hadoop under my existing account and root, which meant localhost SSH connections were refused until I properly installed and started the OpenSSH server daemon (`ssh.service`). Once I created the `hadoop` user, I found that the files were in the wrong user directory, so I re-extracted Hadoop directly inside `/home/hadoop/Desktop/hadoop` to ensure clean folder ownership and permissions. I also verified I could easily access this user remotely over my Tailscale network.

Step 5:

I added all environment variables to `~/ .bashrc` and configured the cluster files (`hadoop-env.sh`, `core-site.xml`, `hdfs-site.xml`, `mapred-site.xml`, and `yarn-site.xml`).

First Issue (Java Version Mismatch): When I attempted to format the NameNode with `hdfs namenode -format`, it failed with an `UnsupportedClassVersionError` (class file version 61.0 vs 52.0). I discovered that the latest Hadoop 3.5.0 release was compiled using Java 17, while the lab guide originally specified Java 8. To resolve this, I installed `openjdk-17-jdk` and updated `JAVA_HOME` in both `~/ .bashrc` and `hadoop-env.sh` to point to `/usr/lib/jvm/java-17-openjdk-amd64`. Re-running `hdfs namenode -format` then succeeded.

```
hadoop@ograzier:~$ jps
42433 SecondaryNameNode
44484 NodeManager
42152 NameNode
42265 DataNode
73085 Jps
44366 ResourceManager
```

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Nameservice Information

Datanode Information

About the Cluster

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HadoopOverviewDatanodesDatanode Volume FailuresSnapshotStartup ProgressUtilities

Overview 'localhost:9000' (-active)

Started:	Thu Sep 17 10:36:11 -0400 2026
Version:	3.5.0, rbbcc7bd797100e6b2c0b4848b5a5193a59a0
Completed:	Tue Mar 24 12:56:00 -0400 2026 by ocauroth from branch-3.5.0
Cluster ID:	CID-ca6c1d95-d871-4e46-b8b5-4d3855713558
Block Pool ID:	BP-1299028136-127.0.1.1-1789655707314

Summary

Security is off.
Safemode is off.
1 files and directories, 0 blocks (0 replicated blocks, 0 ensemble coded block groups) = 1 total filesystem object(s).
Heap Memory used 70.69 MB of 92 MB Heap Memory. Max Heap Memory is 756 MB.
Non Heap Memory used 79.01 MB of 82.63 MB Committed Non Heap Memory. Max Non Heap Memory is 'unbounded'.
Configured Capacity: 232.64 GB
Configured Remote Capacity: 0 B
DFS Used: 28 KB (0%)
Non DFS Used: 64.93 GB
DFS Remaining: 196.22 GB (87.10%)
Block Pool Used: 28 KB (0%)
BlockNodes usage% (Min/Median/Max/stdDev): 0.00% / 0.00% / 0.00% / 0.00%
Live Nodes: 1 (Decommissioned: 0, In Maintenance: 0)
Dead Nodes: 0 (Decommissioned: 0, In Maintenance: 0)
Decommissioning Nodes: 0
Entering Maintenance Nodes: 0
Total Datanode Volume Failures: 0 (0 B)
Number of Under-Replicated Blocks: 0
Number of Blocks Pending Deletion (including replicas): 0
Block Deletion Start Time: Thu Sep 17 10:36:11 -0400 2026
Last Checkpoint Time: Sun Sep 20 14:37:54 -0400 2026
Last HA Transition Time: Never
Enabled Erasure Coding Policies: EC-6-5-1024K

100.76.121.7/870/dhadoop-hadoop-overview

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Nameservice Information

Datanode Information

About the Cluster

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HadoopOverviewUtilities

DataNode on ograzier:9866

Cluster ID:	CID-ca6c1d95-d871-4e46-b8b5-4d3855713558
Started:	Thu Sep 17 10:36:15 -0400 2026
Version:	3.5.0, rbbcc7bd797100e6b2c0b4848b5a5193a59a0

Block Pools

Nameservice Address	Nameservice HA State	Block Pool ID	Actor State	Last Heartbeat Sent	Last Heartbeat Response	Last Block Report	Last Block Report Size (Max Size)
localhost:9000	active	BP-1299028136-127.0.1.1-1789655707314	RUNNING	2s	2s	an hour	0 B (128 MB)

Volume Information

Directory	StorageType	Capacity Used	Capacity Left	Capacity Reserved	Reserved Space for Replicas	Blocks
/home/hadoop/Desktop/hadoop/grh/datanode	DISK	28 KB	196.22 GB	0 B	0 B	0

Hadoop, 2026

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Nameservice Information

Datanode Information

About the Cluster

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HadoopOverviewUtilities

DataNode on ograzier:9866

Cluster ID:	CID-ca6c1d95-d871-4e46-b8b5-4d3855713558
Started:	Thu Sep 17 10:36:15 -0400 2026
Version:	3.5.0, rbbcc7bd797100e6b2c0b4848b5a5193a59a0

Block Pools

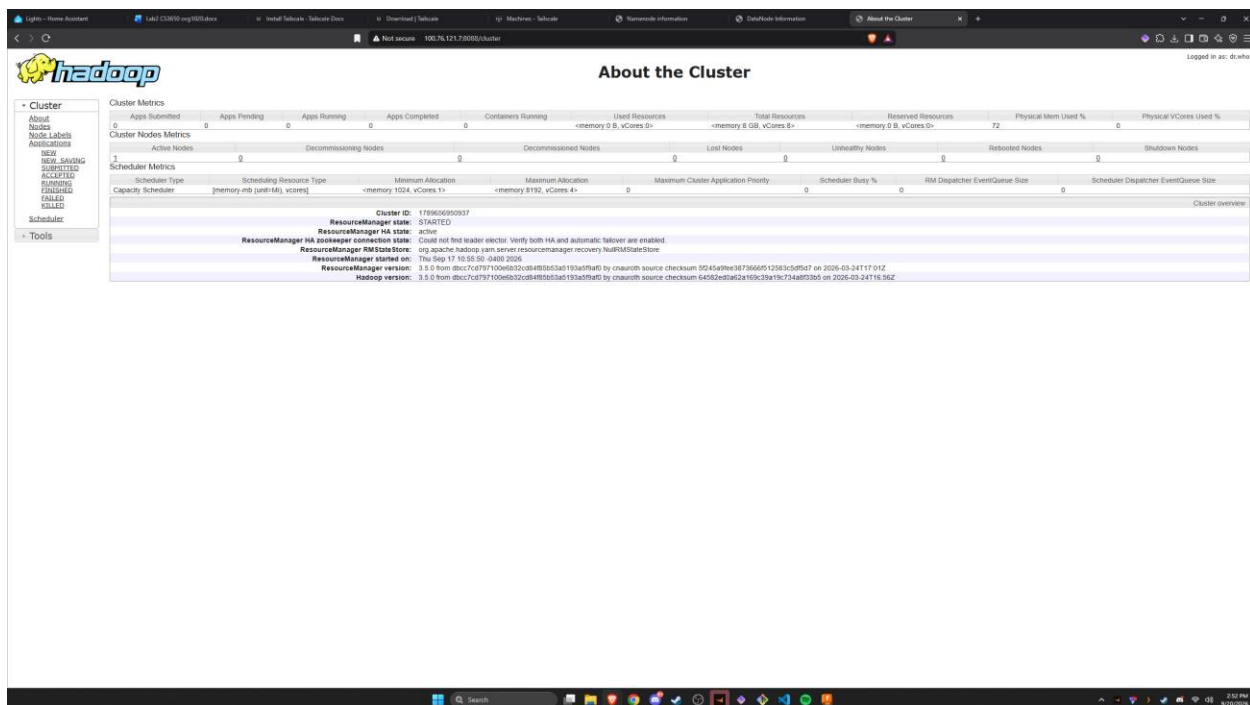
Nameservice Address	Nameservice HA State	Block Pool ID	Actor State	Last Heartbeat Sent	Last Heartbeat Response	Last Block Report	Last Block Report Size (Max Size)
localhost:9000	active	BP-1299028136-127.0.1.1-1789655707314	RUNNING	2s	2s	an hour	0 B (128 MB)

Volume Information

Directory	StorageType	Capacity Used	Capacity Left	Capacity Reserved	Reserved Space for Replicas	Blocks
/home/hadoop/Desktop/hadoop/grh/datanode	DISK	28 KB	196.22 GB	0 B	0 B	0

Hadoop, 2026

100.76.121.7/8704/datanode.html



Second Issue (YARN Web Interface Binding): After starting HDFS (`start-dfs.sh`) and YARN (`start-yarn.sh`), all five daemons were running in `jps`. The HDFS dashboards on ports 9870 and 9864 opened over my Tailscale IP, but port 8088 refused to connect. When I initially changed the hostname to `0.0.0.0`, `start-yarn.sh` failed to launch the ResourceManager daemon (Starting resourcemangers on []). To fix this, I set `yarn.resourcemanager.hostname` back to `127.0.0.1` and explicitly added `yarn.resourcemanager.webapp.address` set to `0.0.0.0:8088` in `yarn-site.xml`.

After restarting YARN, `jps` showed all five daemons running (NameNode, DataNode, SecondaryNameNode, ResourceManager, and NodeManager), and I was able to successfully view all three web interfaces (9870, 9864, and 8088) in my browser.